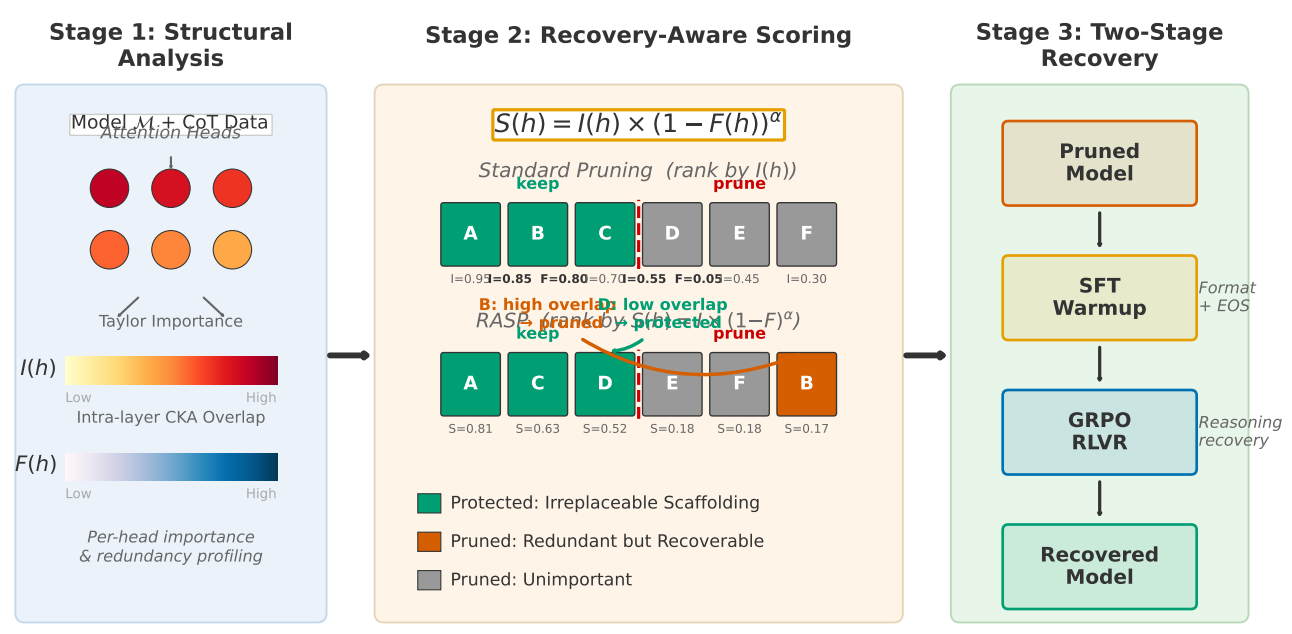

Stage 1: Structural Analysis
Model $\mathcal{M}$ + CoT Data
Attention Heads
Taylor Importance
$I(h)$
Low
High
Intra-layer CKA Overlap
$F(h)$
Low
High
Per-head importance & redundancy profiling
Stage 2: Recovery-Aware Scoring
$S(h) = I(h) \times (1 - F(h))^{\alpha}$
Standard Pruning (rank by $I(h)$)
keep
prune
A
B
C
D
E
F
I=0.95
I=0.85
F=0.80
I=0.70
I=0.55
F=0.05
I=0.45
I=0.30
B: high overlap
D: low overlap
RASP (rank by $S(h) = \ldots \times (1-F)^{\alpha}$)
pruned
protected
keep
prune
A
C
D
E
F
B
S=0.81
S=0.63
S=0.52
S=0.18
S=0.18
S=0.17
Protected: Irreplaceable Scaffolding
Pruned: Redundant but Recoverable
Pruned: Unimportant
Stage 3: Two-Stage Recovery
Pruned Model
SFT Warmup
Format + EOS
GRPO RLVR
Reasoning recovery
Recovered Model